MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3850 REAGENTS – LIME SLAKING EQUIPMENT
LIME SLAKER TANK FOUNDATION 3850-MSC-002

Document Number: **MD-216-3850-EG-CV-CAL-1432**

Revision: **D01**

Revision Date: **30.11.2023**

Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AA03	CED01	38506	OU100695877	2
L&T Document Number:	O19045	450	C	K	24166	003	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	10.10.2020	FIRST ISSUE	SUE	BDA	AG	
C01	02.11.2020	ISSUED FOR CONSTRUCTION	SUE	BDA	AG	
D01	30.11.2023	AS BUILT	SUE	BDA	AG	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3850-MSC-002 LIME SLAKER TANK**
The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in standard excel sheet. Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation Bending moment & Shear Force values are calculate in the conventional method & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Wall	50
	Slab on Grade	20

5. LOAD

5.1 Primary Loads

3850-MSC-002 LIME SLAKER TANK foundation design the primary load values are taken from **Attachment - 1 Mixing Tank General Arrangement** drawing

1 DEAD LOAD (D)
2 LIVE LOAD (L)
3 SEISMIC LOAD (S)
4 WIND LOAD (W)
5 FLUID LOAD (F)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	D + F
102	D + F + L
103	D + F + W
104	D + F + 0.7*E
105	D + F + 0.75*W + 0.75*L
106	D + F + 0.75*(0.7*E) + 0.75*L
107	0.6*D + W
108	0.6*D + 0.7*E

5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	1.4*(D + F)
202	1.4*(D + F) + 1.7*L
203	1.2*D + 0.8*W
204	1.2*D + 1.6*W + 0.5*L
205	1.2*D + E + 0.5*L
206	0.9*D + 1.6*W
207	0.9*D + E

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6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design factor of safety against overturning is consider 2.0 for permanent load.
For foundation design factor of safety against overturning is consider 1.5 for temporary load.

6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

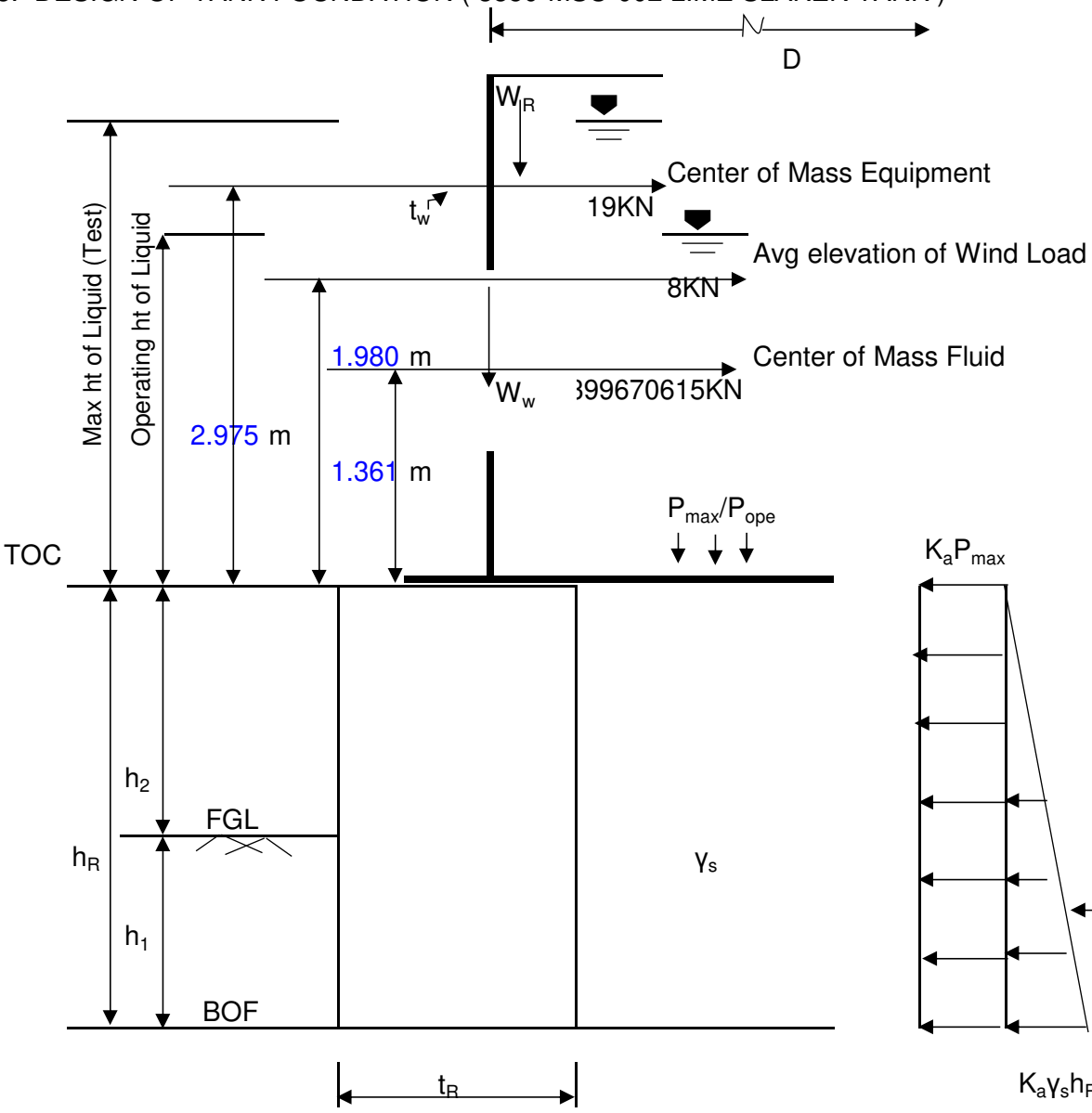
The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m2 at 1.2m depth below FGL
The neareast Bore Hole of 3850-REA-001 MILK OF LIME DOSING TANK is BH-12 to BH-15.
The net safe bearing capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is 1200 KN/m2 at 1.5m depth below FGL.

However for safe design purpose we have considered the Net Safe Bearing Capacity at founding level as **670 KN/m2**.

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8. DESIGN OF TANK FOUNDATION (3850-MSC-002 LIME SLAKER TANK)



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	3.000	m
Self weight of tank =	17	KN
Self weight of tank per m of circumference, W_T =	1.8	KN/m
Live Load =	11.3	KN
Live load per m of circumference, W_L =	1.2	KN/m
Maximum weight of Fluid (F) =	182.0	KN
Pressure at base of tank , P_{max} = F / A =	25.8	KN/m^2
Operating weight of Slurry ($F_{operating}$) =	0	KN
Pressure at base of tank , $P_{operating}$ = ($F_{operating}$) / A =	0	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete, γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil, γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all} =	670	KN/m^2	
Coefficient of active earth pressure, K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	3.2	m
FGL to Bottom of foundation , h_1 =	1.5	m
FGL to Top of concrete , h_2 =	1.7	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall, D_i =	2.6	m
Outer dia of ringwall, D_o =	3.5	m
$S_{xx} = \pi(D_o^4 - D_i^4)/32D_o$ =	2.93	m^3

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Check of soil bearing pressure (Max weight of Slurry)

Wind force = 8 KN acting at a height of 5.180 m from BOF

M_{OT} = Overturning moment = 41.4 KN-m

M_{OT} / S_{xx} = 14.16 KN/m^2

Load Combination : 102. D + F + L

$q_{RF} = (W_{T+} W_L)/t_R + P_{max} + h_R\gamma_c$ = 109.2 KN/m^2 **< 670KN/m2, O.K.**

Load Combination : 103. D + F ± W

$q_{max} = (W_T)/t_R + P_{max} + h_R\gamma_c + M_{OT}/S_{xx}$ = 120.7 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = (W_T)/t_R + P_{max} + h_R\gamma_c - M_{OT}/S_{xx}$ = 92.4 KN/m^2 **No (-)ve pressure**

Load Combination : 105. D + F ± 0.75*W + 0.75*L

$q_{max} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c + 0.75*M_{OT}/S_{xx}$ = 119.2 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = (W_T+0.75*W_L)/t_R + P_{max} + h_R\gamma_c - 0.75*M_{OT}/S_{xx}$ = 97.94 KN/m^2 **No (-)ve pressure**

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 107. 0.6*D ± W

$q_{RE} = (W_T)/t_R + h_R\gamma_c$ = 80.8 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 62.6 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 34.32 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 41.4 KN-m

M_{OR} = Restoring moment = (16.9646003293849+PI()*3*0.45*3.2*24)*3/2 = 514 KN-m

FOS = M_{OR} / M_{OT} = 514.03 / 41.44 = 12.40 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating weight of Fluid):

Seismic force at center of mass equipment = 19 KN
acting at a height of 6.175 m from BOF

M_{OT} = Overturning moment = 117.3 KN-m

M_{OT} / S_{xx} = 40.08 KN/m^2

Load Combination : 101. D + F

$q_{RF} = (W_T)/t_R + F + h_R\gamma_c$ = 106.6 KN/m^2 **< 670KN/m2, O.K.**

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Load Combination : 107. D + F ± 0.7*E

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 134.6 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 78.5 KN/m^2 **No (-)ve pressure**

Load Combination : 108. 0.6*D ± 0.7*E

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 76.53 KN/m^2 **< 670KN/m2, O.K.**

$q_{max} = 0.6*(W_T)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = 20.43 KN/m^2 **No (-)ve pressure**

Check for Overturning (Empty Tank): For Seismic Load

M_{OT} = Overturning moment = 117.3 KN-m

M_{OR} = Restoring moment = (16.9646003293849+PI()*3*0.45*3.2*24)*3/2 = 514 KN-m

FOS = M_{OR} / M_{OT} = 514.03 / 117.325 = 4.38 **> 1.5, Hence O.K.**

Ringwall Reinforcement :

Radial outward force on ringwall, $R_{F1} = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 59.3 KN/m

For Hydrotest :

Radial outward force on ringwall, R_{F2}= $K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 63.3 Kn/m

Hoop tension in ringwall, T = 1/2*Max(R_{F1},R_{F2})*D = 94.88 KN

Factored design tension, T_u= 132.8 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 351.4 mm^2 ϕ = 0.9

$A_{st,min}$ = 0.002* $h_R t_R$ = 2880 mm^2

$A_{st,reqd}$ (per face of ring wall)= 1440 mm^2

Required : 12 dia bar @ 251.4 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 1810 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min}$ = 0.0012* t_R = 540 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291.1 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 392.9 mm^2 **O.K.**

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ATTACHMENT - 1 : Mechanical Input Drawing.

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ITEM	DESCRIPTION	MATERIAL	DOCUMENT NUMBER
1	MIXING TANK	MILD STEEL, EN 10025 S 355 JR-AR,	001-1878-000-004-010
2	AGITATOR	MILD STEEL, EN 10025 S 355 JR-AR,	001-1878-000-000-108
3	WET SCRUBBER	MILD STEEL, EN 10025 S 355 JR-AR,	001-1878-000-000-104
4	HANDRAIL ASSEMBLY	MILD STEEL, EN 10025 S 355 JR-AR,	001-1878-000-004-118

ELECTRICAL DATA

1.	MOTOR VOLTAGE	400VAC
2.	ELECTRIC MOTORS	ABB
3.	FIELD POWERED INSTRUMENTS	24 VDC
4.	ANALOGUE INSTRUMENTS & CONTROL VOLTAGE	24VDC / 4-20mA HART
5.	SOLENOIDS	24VDC
6.	DIGITAL INSTRUMENTS I/O	24VDC
7.	INSTRUMENTATION	E&H (HART) , VEGA (HART)

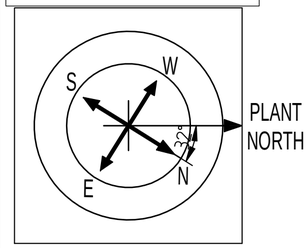
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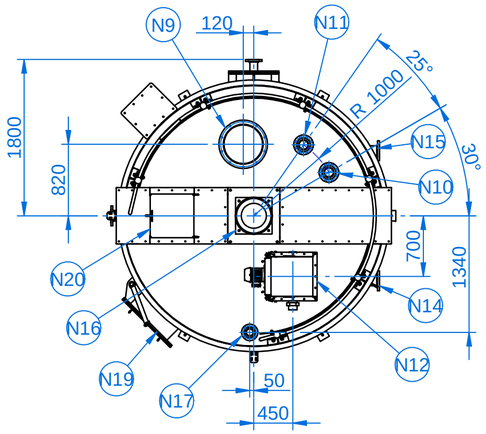
AMBIENT TEMPERATURE

WIND SPEED
- NON PROCESS AREA 49 m.s^{-1}
- PROCESS AREA 43 m.s^{-1}

DESIGN SEISMIC LOADING
- NON PROCESS AREA 0.02 g
- PROCESS AREA 0.015 g

DESIGN TEMPERATURE 110 °C





PLAN VIEW

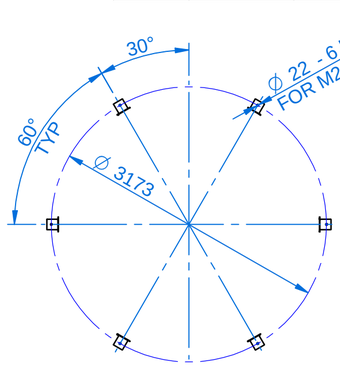
FOUNDATION LOADS

NORMAL FOUNDATION LOADS

DESCRIPTION		DATA	
Dead Load per Unit Length (Shell and Roof)	[kN/m]	DL =	1,80
Live Load per Unit Length (Live Load on Roof)	[kN/m]	LL =	1,20
Total Dead Weight (Shell, Roof and Liquid)	[kN]	Wt =	199
WIND LOADS			
Overturning Moment due to Wind	[kNm]	Mw =	11
Base Shear Force due to Wind	[kN]	Vw =	8
SEISMIC LOADS			
Base Shear due to Seismic Load	[kN]	Vs =	19
Overturning Moment - Ring Wall	[kNm]	Mrw =	27
Overturning Moment - Slab	[kNm]	Ms =	83
Vertical Seismic Force (Shell and Roof)	[kN]	Fsv =	2
Vertical Seismic Force (Liquid)	[kN]	Fsliq =	18

NOZZLE SCHEDULE

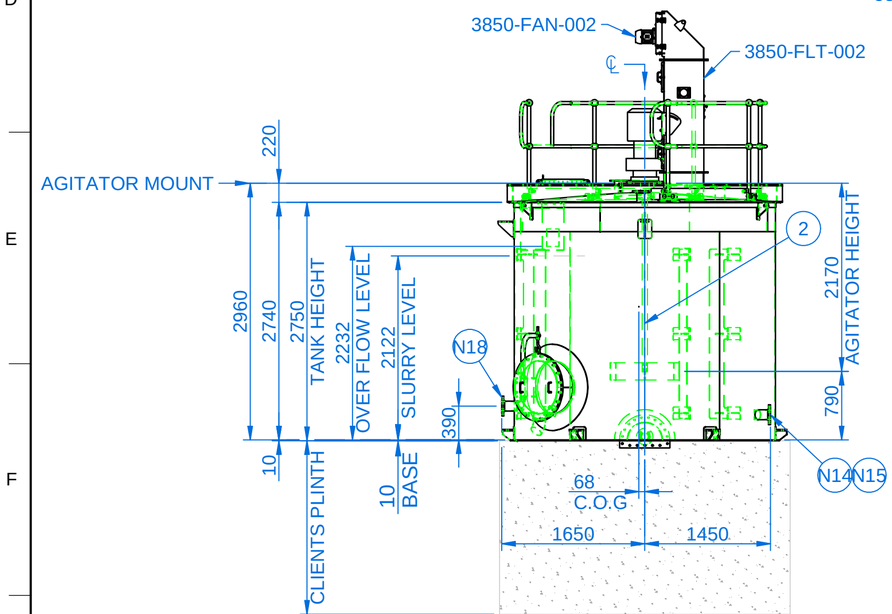
NOZZLE #	TAG #	DESCRIPTION	MATERIAL	CONNECTION DETAILS
9	N/A	SLAKER LIME INLET	MILD STEEL, EN 10025 S 355 JR-AR,	BSM 450NB FLANGE DRILLING
10	3850-LIT-85522	RADAR LEVEL TRANSMITTER	MILD STEEL, EN 10025 S 355 JR-AR,	100NB, 4" ANSI B16.5 150LB SO RF
11	N/A	BLIND	MILD STEEL, EN 10025 S 355 JR-AR,	100NB, 4" ANSI B16.5 150LB SO RF
12	N/A	WET SCRUBBER	MILD STEEL, EN 10025 S 355 JR-AR,	560mm X 560mm
13	N/A	SLAKER COMPARTMENT DRAIN	MILD STEEL, EN 10025 S 355 JR-AR,	80NB, 3" ANSI B16.5 150LB SO RF
14	N/A	SLAKER OUTLET-1	MILD STEEL, EN 10025 S 355 JR-AR,	100NB, 4" ANSI B16.5 150LB SO RF
15	N/A	SLAKER OUTLET-2	MILD STEEL, EN 10025 S 355 JR-AR,	100NB, 4" ANSI B16.5 150LB SO RF
16	N/A	AGITATOR CONNECTION	MILD STEEL, EN 10025 S 355 JR-AR,	420mm X 420mm
17	N/A	BY PASS CONNECTION	MILD STEEL, EN 10025 S 355 JR-AR,	80NB, 3" ANSI B16.5 150LB SO RF
18	3850-TT-85523	TEMPERATURE TRANSMITTER	MILD STEEL, EN 10025 S 355 JR-AR,	100NB, 4" ANSI B16.5 150LB SO RF
19	M1	MAN HOLE	MILD STEEL, EN 10025 S 355 JR-AR,	600NB, 24" ANSI B16.5 FLANGE DRILLING
20	M2	INSPECTION HATCH	MILD STEEL, EN 10025 S 355 JR-AR,	500mm X 500mm
21	N/A	OVER FLOW	MILD STEEL, EN 10025 S 355 JR-AR,	125NB, 5" ASTM A106 GRADE B



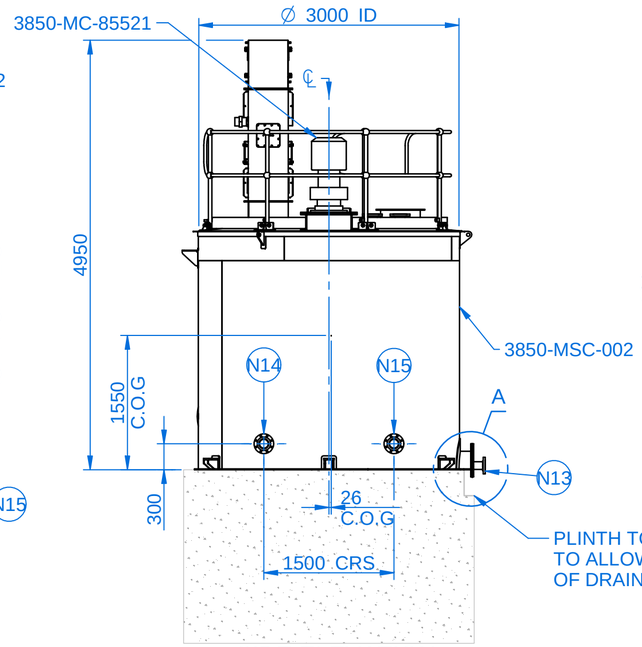
TANK MOUNT DETAILS
1:50

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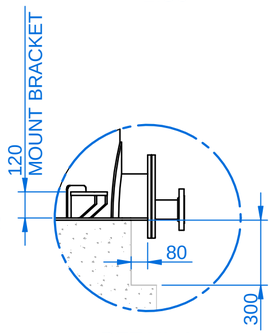
CERTIFIED DRAWING



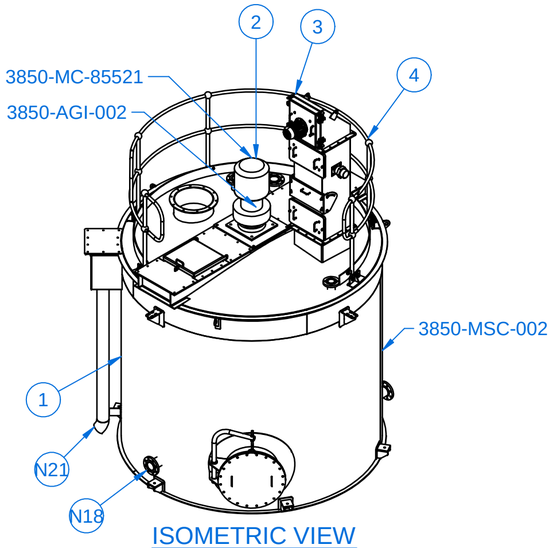
REAR ELEVATION



SIDE ELEVATION



DETAIL A
PLINTH NOTCH DETAIL
1:20



ISOMETRIC VIEW

NOTE
Please note REV C changes done during tank fabrication.

MASS/ITEM: 3321 kg

REV	DESCRIPTION	DATE	NAME	CHECKED
0	CERTIFIED FOR SUBMISSION	25/09/20	Z.G.	R.A.B
F	NOZZLE 11 BLIND WAS SPARE	07/09/20	P.C.	R.A.B
E	MASS & NOZZLE NUMBERS ADDED	27/08/20	A.K.	R.A.B
D	UPDATED INLET SPIGOT POSITION	03/08/20	M.NEL	R.A.B
C	EQUIPMENT ORIENTATION CHANGED AS PER CLIENTS REQUEST	24/07/20	Z.G.	R.A.B
B	GENERAL NOTES UPDATED	10/06/20	M.NEL	R.A.B.
A	ISSUED FOR APPROVAL	30/03/20	R.A.B.	R.A.B.

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PROJECT:	LIME SLAKING PLANT
CUSTOMER:	OUTOTEC
JOB No.	EJ1878
DWG TITLE	GENERAL ARRANGEMENT LIME SLAKER (3850-MSC-002) MIXING TANK
DRAWN:	R.A.B.
CHECKED:	R.A.B.
DATE	11/09/20
DWG NUMBER:	001-1878-000-000-103
SCALE:	1:50
REV	0
THIRD ANGLE	
SIZE	A2